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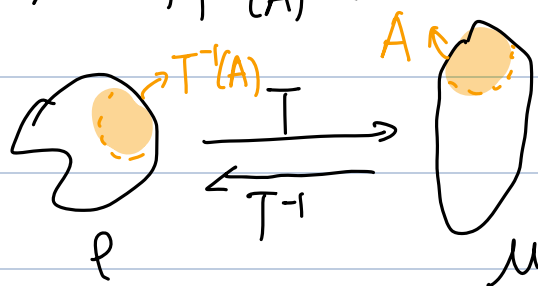
- Optimal Transport problem (Monge Formulation)

Given two densities of mass (probability measures), $\rho, \mu \geq 0$ on $\mathbb{R}^d$

$$\int \rho(x) dx = \int \mu(y) dy = 1,$$

find a map $T: \mathbb{R}^d \rightarrow \mathbb{R}^d$ s.t. $T\#\rho = \mu$ (push-forward constraint)

ie. $\int_A \mu(y) dy = \int_{T^{-1}(A)} \rho(x) dx$ for any Borel subset $A \subset \mathbb{R}^d$



and minimizing transportation cost

$$M(T) := \int_{\mathbb{R}^d} c(x, T(x)) \rho(x) dx$$

where $c(x, y)$ is some "cost"

-Eg. $c(x, y) = \frac{1}{2} \|x - y\|^2$

- Density Estimation (using push forward constraint)

- measure ρ on X , measure μ on Y

- $T\#\rho \Rightarrow \mu(A) = (T\#\rho)(A) = \rho(T^{-1}(A))$

require $T\#\rho = \mu$: $\mathbb{E}_{T\#\rho}[\phi] = \mathbb{E}_{\rho}[\phi \circ T] = \mathbb{E}_{\mu}[\phi], \forall$ measurable

$$\int (\phi \circ T)(x) \rho(x) dx = \int \phi(y) \mu(y) dy$$

func. ϕ

$$\int \phi(T(x)) p(x) dx = \int \phi(y) \mu(y) dy$$

Change of variable ($y = T(x)$)

$$= \int \phi(T(x)) \mu(T(x)) |T'(x)| dx$$



abs val of det of Jacobian matrix $T'(x)$

$$\bullet \quad p(x) = \mu(T(x)) |T'(x)|$$

$$\begin{aligned} y &= T(x) \\ x &= T^{-1}(y) \Rightarrow p(T^{-1}(y)) = \mu(y) \frac{1}{|T^{-1}'(y)|} \\ &\quad \downarrow \\ &\text{Jacobian of } T^{-1} \end{aligned}$$

$$\begin{aligned} &\text{Jacobian matrix:} \\ &X = [x_1, \dots, x_d]^T \in \mathbb{R}^d \\ &T: \mathbb{R}^d \rightarrow \mathbb{R}^d = \begin{pmatrix} T_1(x_1, \dots, x_d) \\ \vdots \\ T_d(x_1, \dots, x_d) \end{pmatrix} \\ &T'(x) = \begin{bmatrix} \frac{\partial T_1}{\partial x_1} & \frac{\partial T_1}{\partial x_2} & \dots & \frac{\partial T_1}{\partial x_d} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial T_d}{\partial x_1} & \dots & \dots & \frac{\partial T_d}{\partial x_d} \end{bmatrix} \end{aligned}$$

• Monte Carlo Integration

identically independent distributed
↓

a R.V. $X \sim p(x)$ PDF (density), $X_1, \dots, X_N$ iid samples $\sim p$

$$\bullet \quad \mathbb{E}[\phi] = \mathbb{E}_p[\phi(x)] = \int \phi(x) p(x) dx$$

$$\approx \frac{1}{N} \sum_{i=1}^N \phi(x_i) \quad (\text{sample average})$$

(CLT, Law of Large # $\Rightarrow$ convergence theorem)

• Importance Sampling: $X_1, \dots, X_N \sim \tilde{p}$ (referenced density)

$$\frac{p}{\tilde{p}}(x) = \frac{p(x)}{\tilde{p}(x)} \text{ known}$$

$$\mathbb{E}_p[\phi(x)] = \int \phi(x) \frac{p(x)}{\tilde{p}(x)} \tilde{p}(x) dx = \mathbb{E}_{\tilde{p}}[\phi(x) \cdot \frac{p}{\tilde{p}}(x)]$$

$$(\text{weighted sum}) \approx \frac{1}{N} \sum_{i=1}^N \phi(x_i) \cdot \frac{p}{\tilde{p}}(x_i)$$

weight

• Gradient Flow / Gradient Descent

• ODE

$x(t)$
 ↑
 position/displacement

$$(\text{velocity}) \dot{x} = \frac{dx}{dt} = V(x(t))$$

$$x(t) = x(t_0) + \int_{t_0}^t V(x(s)) ds$$

$$\text{— eg. } V(x) = V \text{ const.} \Rightarrow x(t) = x(t_0) + V(t - t_0)$$

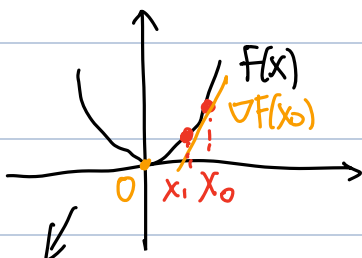
• Gradient Flow

$$\left(\frac{x(t_{k+1}) - x(t_k)}{\Delta t} \approx \right) \dot{x} = \frac{dx}{dt} = -\nabla_x F(x(t)) = -\nabla_x \overset{\substack{\text{some objective to minimize}}}{F}(x)$$

$$\Rightarrow \dot{x} = 0 \text{ (velocity} = 0) \Rightarrow \nabla_x F = 0$$

• Gradient Descent (discretize flow)

$$f(x) = x^2$$



f is convex!

$$x^{t+1} = x^t - \eta \cdot \nabla F(x^t)$$

↳ learning rate

$$-\nabla f(x) = -2x$$

• Normalizing Flow

$$\frac{dT(x,t)}{dt} = - \left. \frac{d \text{KL}(\emptyset \circ T(\cdot, t))}{d\emptyset} \right|_{\emptyset = \text{id}}$$

$$T(\cdot, t^n) = \emptyset^n \circ \emptyset^{n-1} \circ \dots \circ \emptyset^1$$

$$\emptyset^k = \emptyset(\cdot, \theta^k) \quad , \quad \text{assume } \emptyset(\cdot, \theta_0) = \text{id}.$$

$$\text{KL}(T(\cdot, t)) = \int \log \left(\frac{p(x)}{\tilde{p}_t(x)} \right) p(x) dx \quad (\text{"difference btw } p \text{ \& } \tilde{p}_t")$$

$$\rightarrow T(\cdot, t) \# p \sim \tilde{p}_t$$

$$\text{KL}_t = \text{KL}(\emptyset \circ T(\cdot, t))$$

$$\tilde{p}_t^\emptyset \sim \emptyset \circ T(\cdot, t) \# p$$

The symbol after the # is always the source distribution — the one from which you are sampling before applying the transformation

$$\emptyset(T(\cdot, t), \theta_t) \sim \tilde{p}_t^\emptyset$$

$$\dot{\theta}_t = - \frac{\partial \text{KL}_t}{\partial \theta_t}$$

$$\theta^{n+1} = \theta_0 - \frac{\partial \text{KL}_t}{\partial \theta_t} \Big|_{\theta^n}$$

theta_n

$$\downarrow \quad \emptyset^{n+1} = \emptyset(\cdot, \theta^{n+1})$$

